> 🛠️ **2026.10 수정본 (v11)** — 원본: ch02_03_02B_titanic_lgbm_analysis.ipynb
>
> LightGBM 4.x부터 `fit()`에서 `early_stopping_rounds`, `verbose` 인자가 빠졌어요. 그래서 `callbacks=[lgb.early_stopping(100)]` 방식으로 바꿨습니다. 나머지 흐름은 원본 그대로예요.

타이타닉 생존자 예측 대회

학습 목표

- LightGBM 모델을 구축해 본다.
- EDA과정을 줄이고 모델을 구축하고 평가하고 개선시키는 것에 목적을 둔다.
- 매개변수를 변경하여 최적의 모델을 찾는다.
- 변수의 결측치 처리를 하여 모델의 성능을 향상시켜본다.

학습 내용

- 타이타닉 데이터 셋을 활용한 머신러닝 모델 구축
- 최적의 매개변수를 찾기
- 결측치 처리 후, 추가 특징을 추가하기

데이터 분석 목표

- 미션 : Titanic의 생존자를 예측해라
- 미션 내용 및 데이터
  - https://www.kaggle.com/c/titanic

머신러닝 구축 및 개선 전반적인 절차

- 1. 무엇을 하는가? 생존자를 예측하라.
- 2. 데이터 수집 - 내부, 외부...수집
- 3. 데이터 탐색 및 전처리
- 4. 문제 인식 - 머신러닝 어떤 문제? 지도학습/비지도학습 - 회귀/분류
- 5. 머신러닝 모델 선택
- 6. 모델 학습 및 평가
- 7. 최종 모델로 test의 생존 유무 예측
- 8. 모델 개선

In [1]:
```
### 02. 데이터 수집 – 내부, 외부...수집
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns

train = pd.read_csv("./data/titanic/train.csv")
test = pd.read_csv("./data/titanic/test.csv")
sub = pd.read_csv("./data/titanic/gender_submission.csv")

train.shape, test.shape, sub.shape
```

Out[1]: ((891, 12), (418, 11), (418, 2))

In [2]:
```
### 03. 데이터 탐색 및 전처리
### head(), tail(), columns, shape, info(), describe(), dtype
### barplot, histogram, boxplot, scatter, corr()..
print( train.columns )
print( test.columns )
print( sub.columns )
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
Index(['PassengerId', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch',
       'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
Index(['PassengerId', 'Survived'], dtype='str')
```

Data Fields

구분 | 설명 | 값 | -----|-----|-----| **Survival** | 생존 여부 | Survival. 0 = No, 1 = Yes **Pclass** | 티켓의 클래스 |Ticket class. 1 = 1st, 2 = 2nd, 3 = 3rd **Sex** | 성별(Sex) | 남(male)/여(female) **Age** | 나이(Age in years.) | **SibSp** | 함께 탑승한 형제와 배우자의 수 /siblings, spouses aboard the Titanic. | **Parch** | 함께 탑승한 부모, 아이의 수 | # of parents / children aboard the Titanic. **Ticket** | 티켓 번호(Ticket number) | (ex) CA 31352, A/5. 2151 **Fare** | 탑승료(Passenger fare) | **Cabin** | 객실 번호(Cabin number) | **Embarked** | 탑승 항구(Port of Embarkation) | C = Cherbourg, Q = Queenstown, S = Southampton

- siblings : 형제, 자매, 형제, 의붓 형제
- spouses : 남편, 아내 (정부와 약혼자는 무시)
- Parch : Parent(mother, father), child(daughter, son, stepdaughter, stepson)

04~05 문제 확인 및 머신러닝 모델 선택

In [3]:
```
### 04. 문제 확인 - 머신러닝 어떤 문제? 지도학습/비지도학습 - 회귀/분류
### Survived : 생존 유무(0, 1)
### knn, decisionTree, lightGBM
### 05. 머신러닝 모델 선택
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from lightgbm import LGBMClassifier
import lightgbm as lgb  # 2026.10 수정: early_stopping 콜백 사용을 위해 추가
```

In [4]:
```
### 모델을 위한 데이터 컬럼 선택
sel = ['PassengerId', 'Pclass', 'SibSp', 'Parch', 'Fare']
all_X = train[ sel  ]  # 여러개 컬럼 선택(학습을 위한)
all_y = train['Survived']
```

In [5]:
```
### 자체 평가를 위한 데이터 나누기
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(all_X, all_y,
                                                    test_size=0.3,
                                                    random_state=0)


X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

Out[5]:
```
((623, 5), (268, 5), (623,), (268,))
```

06. 모델 학습 및 평가

In [6]:
```
# from sklearn.neighbors import KNeighborsClassifier
# from sklearn.tree import DecisionTreeClassifier
model1 = KNeighborsClassifier()      # 모델 선택
model1.fit(X_train, y_train)         # 모델 학습
print("knn 모델의 정확도(tr) : ", model1.score(X_train, y_train) )
print("knn 모델의 정확도(test) : ", model1.score(X_test, y_test) )

model2 = DecisionTreeClassifier(random_state=0)
model2.fit(X_train, y_train)
print("의사결정트리 모델의 정확도(tr) : ", model2.score(X_train, y_train) )
print("의사결정트리 모델의 정확도(test) : ", model2.score(X_test, y_test) )
```

```
knn 모델의 정확도(tr) :  0.7303370786516854
knn 모델의 정확도(test) :  0.667910447761194
의사결정트리 모델의 정확도(tr) :  1.0
의사결정트리 모델의 정확도(test) :  0.6604477611940298
```

In [7]:
```
model_lgbm = LGBMClassifier(n_estimators= 400)     # 모델 선택
evals = [(X_test, y_test)]

# 모델 학습
model_lgbm.fit(X_train, y_train,
               eval_metric='logloss',
               eval_set=evals,
               callbacks=[lgb.early_stopping(100), lgb.log_evaluation(10)])

print("lgbm 모델의 정확도(tr) : ", model_lgbm.score(X_train, y_train) )
print("lgbm 모델의 정확도(test) : ", model_lgbm.score(X_test, y_test) )
```

```
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000689 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
Training until validation scores don't improve for 100 rounds
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[10]    valid_0's binary_logloss: 0.582662
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[20]    valid_0's binary_logloss: 0.578576
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[30]    valid_0's binary_logloss: 0.583348
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[40]    valid_0's binary_logloss: 0.596
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[50]    valid_0's binary_logloss: 0.604343
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[60]    valid_0's binary_logloss: 0.614651
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[70]    valid_0's binary_logloss: 0.626191
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[80]    valid_0's binary_logloss: 0.632519
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[90]    valid_0's binary_logloss: 0.638812
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[100]   valid_0's binary_logloss: 0.648549
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[110]   valid_0's binary_logloss: 0.658372
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
Early stopping, best iteration is:
[14]    valid_0's binary_logloss: 0.577216
lgbm 모델의 정확도(tr) :  0.7849117174959872
lgbm 모델의 정확도(test) :  0.7201492537313433
```

- 추가 실습
- 평가 지표 추가
- 가장 최적의 k값을 찾기
- 가장 최적의 max_depth를 찾기

In [8]:
```python
from sklearn import metrics

# 모델 평가를 위한 함수
def get_clf_eval_auc(y_test, y_pred=None, pred_proba=None):
  confusion = metrics.confusion_matrix(y_test, y_pred)
  accuracy = metrics.accuracy_score(y_test, y_pred)
  precision = metrics.precision_score(y_test, y_pred)
  recall = metrics.recall_score(y_test, y_pred)
  F1_score = metrics.f1_score(y_test, y_pred)

  AUC = metrics.roc_auc_score(y_test, pred_proba)

  # 평가지표 출력
  print('오차행렬:\n', confusion)
  print('\n정확도: {:.4f}'.format(accuracy))
  print('정밀도: {:.4f}'.format(precision))
  print('재현율: {:.4f}'.format(recall))
  print('F1: {:.4f}'.format(F1_score))
  print('AUC: {:.4f}'.format(AUC))

  return AUC
```

In [9]:
```python
# 모델을 활용한 예측
preds = model_lgbm.predict(X_test)
pred_proba = model_lgbm.predict_proba(X_test)[:, 1]
```

In [10]:
```
get_clf_eval_auc(y_test, preds, pred_proba)
```

```
오차행렬:
 [[140  28]
 [ 47  53]]

정확도: 0.7201
정밀도: 0.6543
재현율: 0.5300
F1: 0.5856
AUC: 0.7330
```

Out[10]: 0.7329761904761906

In [11]:
```
depth = [3,5,7,9,11, 13]
lr_rate = [0.001, 0.01, 0.1 ]

max_auc = 0
good_par = []
for lr in lr_rate:
    for one_depth in depth:

        model_lgbm = LGBMClassifier(n_estimators=100,
                                    max_depth=one_depth,
                                    learning_rate=lr,
                                    random_state=77)

        model_lgbm.fit(X_train, y_train,
                   eval_metric="logloss",
                   eval_set=evals,
                   callbacks=[lgb.early_stopping(100, verbose=False)])  # 20

        evals = [(X_test, y_test)]

        preds = model_lgbm.predict(X_test)
        pred_proba = model_lgbm.predict_proba(X_test)[:, 1]
        get_AUC = get_clf_eval_auc(y_test, preds, pred_proba)

        if max_auc < get_AUC:
            max_auc = get_AUC
            print("개선된 AUC : ", max_auc)
            print(f"learning_rate  : {lr}, max_depth : {one_depth}")
            good_par.append([lr,one_depth,max_auc])

print("최적의 파라미터 기록 : ", good_par)
```

```
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000617 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[168   0]
 [100   0]]

정확도: 0.6269
정밀도: 0.0000
재현율: 0.0000
F1: 0.0000
AUC: 0.7317
개선된 AUC :  0.7316666666666666
learning_rate  : 0.001, max_depth : 3
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000584 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
.../site-packages/sklearn/metrics/_classification.py:1879: UndefinedMetricWa
rning: Precision is ill-defined and being set to 0.0 due to no predicted sam
ples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[168   0]
 [100   0]]

정확도: 0.6269
정밀도: 0.0000
재현율: 0.0000
F1: 0.0000
AUC: 0.7459
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000619 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
.../site-packages/sklearn/metrics/_classification.py:1879: UndefinedMetricWa
rning: Precision is ill-defined and being set to 0.0 due to no predicted sam
ples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
.../site-packages/sklearn/metrics/_classification.py:1879: UndefinedMetricWa
rning: Precision is ill-defined and being set to 0.0 due to no predicted sam
ples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
.../site-packages/sklearn/metrics/_classification.py:1879: UndefinedMetricWa
rning: Precision is ill-defined and being set to 0.0 due to no predicted sam
ples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

오차행렬:
 [[168   0]
 [100   0]]

정확도: 0.6269
정밀도: 0.0000
재현율: 0.0000
F1: 0.0000
AUC: 0.7428
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000569 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf

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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[157  11]
 [ 60  40]]

정확도: 0.7351
정밀도: 0.7843
재현율: 0.4000
F1: 0.5298
AUC: 0.7401
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000546 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
.../site-packages/sklearn/metrics/_classification.py:1879: UndefinedMetricWa
rning: Precision is ill-defined and being set to 0.0 due to no predicted sam
ples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[148  20]
 [ 55  45]]

정확도: 0.7201
정밀도: 0.6923
재현율: 0.4500
F1: 0.5455
AUC: 0.7338
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000617 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[143  25]
 [ 52  48]]

정확도: 0.7127
정밀도: 0.6575
재현율: 0.4800
F1: 0.5549
AUC: 0.7312
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000601 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[143  25]
 [ 52  48]]

정확도: 0.7127
정밀도: 0.6575
재현율: 0.4800
F1: 0.5549
AUC: 0.7312
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000602 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[143  25]
 [ 52  48]]

정확도: 0.7127
정밀도: 0.6575
재현율: 0.4800
F1: 0.5549
AUC: 0.7312
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000572 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
```

오차행렬:
 [[140 28]
 [49 51]]

정확도: 0.7127
정밀도: 0.6456
재현율: 0.5100
F1: 0.5698
AUC: 0.7647
개선된 AUC : 0.764732142857143
learning_rate : 0.1, max_depth : 3
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000574 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[134  34]
 [ 45  55]]

정확도: 0.7052
정밀도: 0.6180
재현율: 0.5500
F1: 0.5820
AUC: 0.7393
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000600 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[136  32]
 [ 48  52]]

정확도: 0.7015
정밀도: 0.6190
재현율: 0.5200
F1: 0.5652
AUC: 0.7385
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000588 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
오차행렬:
 [[140  28]
 [ 47  53]]

정확도: 0.7201
정밀도: 0.6543
재현율: 0.5300
F1: 0.5856
AUC: 0.7330
최적의 파라미터 기록 :  [[0.001, 3, 0.7316666666666666], [0.001, 5, 0.748928571428
5714], [0.1, 3, 0.764732142857143]]
```

07. 최종 모델로 test의 생존 유무 예측

In [12]:
```
# test 데이터 셋의 컬럼 선택
# sel = ['PassengerId', 'Pclass', 'SibSp', 'Parch', 'Fare']

last_test = test[sel]
last_test.isnull().sum()
```

Out[12]:
```
PassengerId    0
Pclass         0
SibSp          0
Parch          0
Fare           1
dtype: int64
```

In [13]:
```
model_last_lgbm = LGBMClassifier(n_estimators=100,
                                 max_depth=3,
                                 learning_rate=0.1,
                                 random_state=77)
```

In [14]:
```
### 최종 모델 - lightGBM은 결측치 처리를 해 줌.
model_last_lgbm.fit(X_train, y_train)
pred = model_last_lgbm.predict(last_test)
pred
```

```
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000676 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 332
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 5
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

Out[14]:
```
array([0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0,
       1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0,
       1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1,
       0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0,
       1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 0,
       0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1,
       1, 1, 1, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0,
       0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0,
       1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1,
       1, 0, 0, 1, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
       0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0,
       0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0,
       1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0,
       1, 1, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0,
       1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0,
       1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1,
       0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1])
```

In [15]:
```
sub['Survived'] = pred
sub.to_csv("lgbm_baseline.csv", index=False)
```

08. 모델 개선

- Sex를 우리 학습 모델에 추가해보기. (male, female)를 숫자로 변경하기

In [16]:
```
train['Sex'].unique()
```

Out[16]:
```
<StringArray>
['male', 'female']
Length: 2, dtype: str
```

In [17]:
```
train['Sex'].value_counts()
```

Out[17]:
```
Sex
male      577
female    314
Name: count, dtype: int64
```

In [18]:
```
train.loc[ train['Sex']=='male' , 'Sex_num' ] = 0
train.loc[ train['Sex']=='female' , 'Sex_num' ] = 1

train.head(7)
```

Out[18]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	7
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	5
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	5

실습

- test 셋의 컬럼을 변경해 보기

In [19]:

```
test.loc[ test['Sex']=='male' , 'Sex_num' ] = 0
test.loc[ test['Sex']=='female' , 'Sex_num' ] = 1

test.head(7)
```

Out[19]:

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cab
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	Na
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	Na
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	Na
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	Na
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	Na
5	897	3	Svensson, Mr. Johan Cervin	male	14.0	0	0	7538	9.2250	Na
6	898	3	Connolly, Miss. Kate	female	30.0	0	0	330972	7.6292	Na

In [20]:
```
train.columns
```

Out[20]:
```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked', 'Sex_num'],
      dtype='str')
```

In [21]:
```
sel = ['PassengerId', 'Pclass', 'SibSp', 'Parch', 'Fare', 'Sex_num']
all_X = train[ sel  ]  # 여러개 컬럼 선택(학습을 위한)
all_y = train['Survived']


### 데이터 나누기
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(all_X, all_y,
                                                    test_size=0.3,
                                                    random_state=0)

# 마지막 테스트를 위한 데이터
last_test = test[ sel ]
```

In [22]:
```
model1 = KNeighborsClassifier(n_neighbors=22)      # 모델 선택
model1.fit(X_train, y_train)         # 모델 학습
print("knn 모델의 정확도(tr) : ", model1.score(X_train, y_train) )
```

```
print("knn 모델의 정확도(test) : ", model1.score(X_test, y_test) )

model2 = DecisionTreeClassifier(max_depth=5, random_state=0)
model2.fit(X_train, y_train)
print("의사결정트리 모델의 정확도(tr) : ", model2.score(X_train, y_train) )
print("의사결정트리 모델의 정확도(test) : ", model2.score(X_test, y_test) )
```

```
knn 모델의 정확도(tr) :  0.6709470304975923
knn 모델의 정확도(test) :  0.7201492537313433
의사결정트리 모델의 정확도(tr) :  0.8378812199036918
의사결정트리 모델의 정확도(test) :  0.7910447761194029
```

In [23]:

```
model_lgbm = LGBMClassifier(n_estimators= 400)      # 모델 선택
evals = [(X_test, y_test)]

# 모델 학습
model_lgbm = LGBMClassifier(n_estimators=100,
                            max_depth=3,
                            learning_rate=0.1,
                            random_state=77)

model_lgbm.fit(X_train, y_train,
               eval_metric='logloss',
               eval_set=evals,
               callbacks=[lgb.early_stopping(100, verbose=False)])  # 2026.1

print("lgbm 모델의 정확도(tr) : ", model_lgbm.score(X_train, y_train) )
print("lgbm 모델의 정확도(test) : ", model_lgbm.score(X_test, y_test) )
```

```
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000715 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 334
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 6
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
lgbm 모델의 정확도(tr) :  0.8170144462279294
lgbm 모델의 정확도(test) :  0.8171641791044776
.../site-packages/lightgbm/sklearn.py:1106: LGBMDeprecationWarning: The argu
ment 'eval_set' is deprecated, use 'eval_X' and 'eval_y' instead.
  eval_set = _validate_eval_set_Xy(eval_set=eval_set, eval_X=eval_X, eval_y=
eval_y)
```

In [24]:
```
last_test.isnull().sum()
```

Out[24]:
```
PassengerId    0
Pclass         0
SibSp          0
Parch          0
Fare           1
Sex_num        0
dtype: int64
```

In [25]:
```
print(last_test['Fare'].mean())
last_test['Fare'] = last_test['Fare'].fillna(last_test['Fare'].mean())
```

```
35.627188489208635
```

In [26]:
```
### 최종 모델
model2 = DecisionTreeClassifier(max_depth=5, random_state=0)
model2.fit(X_train, y_train)
pred = model2.predict(last_test)

sub['Survived'] = pred
sub.to_csv("second_sex_add.csv", index=False)
```

In [27]:
```
### 최종 모델
model_lgbm = LGBMClassifier(n_estimators=100,
                            max_depth=3,
                            learning_rate=0.1,
                            random_state=77)
model_lgbm.fit(X_train, y_train)
pred = model_lgbm.predict(last_test)

sub['Survived'] = pred
sub.to_csv("lgbm_sex_add.csv", index=False)
```

```
[LightGBM] [Info] Number of positive: 242, number of negative: 381
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of te
sting was 0.000642 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 334
[LightGBM] [Info] Number of data points in the train set: 623, number of use
d features: 6
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.388443 -> initscore=-0.453
862
[LightGBM] [Info] Start training from score -0.453862
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

더 생각해 보고 실습해 보기

- 승선상 정보 추가
- Cabin 정보 추가
- Name 을 이용한 정보 추가
- 결측치 처리 후, Age 정보 추가
- (자율 과제) 이중의 하나의 정보를 추가해서 제출해 보기

내용

- 최종 update : 2023/05/17

In [ ]: